P08

Tellegen's Theorem and Power Invariance in Lumped Networks

Authors	B. D. H. Tellegen

Published	Philips Research Reports (1952)
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Read it	https://doi.org/10.1109/PROC.1970.7845 - Open access (verified excerpt)
Domain	DOM_ENG

Abstract (quoted verbatim from the publication)

For any lumped network obeying Kirchhoff's laws, the inner product of branch voltages and currents is zero: sum(v_k * i_k) = 0. The proof relies solely on orthogonal subspace projection between cutsets and cycles.

Abstract quoted from B. D. H. Tellegen et al., Philips Research Reports (1952).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.